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## Article

**Keywords:** Bearing Health Management, Inventory Management, Deep Reinforcement Learning, Reward Shaping, Joint Decision-Making

**Posted Date:** January 29th, 2026

**DOI:** <https://doi.org/10.21203/rs.3.rs-8614588/v1>

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**Additional Declarations:** No competing interests reported.

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# Risk-Sensitive Joint Inventory-Maintenance Strategy for Bearing Health Management under Prognostic Uncertainty: An Uncertainty-Aware Deep Reinforcement Learning Approach

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**Abstract:** Traditional separation of inventory management and Prognostics and Health Management (PHM) often leads to resource misallocation. While Deep Reinforcement Learning (DRL) offers a promising solution for joint decision-making, standard agents typically treat Prognostics and Health Management predictions as deterministic ground truths. However, in real-world scenarios, RUL predictions inherently contain stochastic errors. Ignoring this uncertainty leads to risk-blind policies that fail to buffer against sudden failures when prediction confidence is low. To address this, this paper proposes an Uncertainty-Aware collaborative adaptive inventory strategy. First, we introduce a Bayesian uncertainty quantification mechanism using Monte Carlo Dropout to estimate not only the RUL value but also its prediction variance. Second, to overcome the agent's myopic behavior, a novel Asymmetric Cost-Aware Reward Shaping mechanism is designed. By strategically decoupling the training and evaluation reward functions—specifically by introducing safety stock penalties and attenuating holding costs during training—the agent is guided to establish robust inventory buffers against supply chain uncertainties. Simulation results demonstrate that the proposed Risk-Sensitive PPO strategy significantly outperforms deterministic baselines, reducing total costs by 40.3% under high-noise environments.

**Keywords:** Bearing Health Management; Inventory Management; Deep Reinforcement Learning; Reward Shaping; Joint Decision-Making

## 1. Introduction

In the evolution of Industry 4.0 and smart manufacturing, the coordinated scheduling of predictive maintenance for equipment and spare parts supply chains has become critical to reducing total lifecycle costs [1]. As core components of high-end equipment, rolling bearings often incur downtime losses far exceeding the value of the spare parts themselves when sudden failures occur. Therefore, how to ensure high system availability while precisely controlling the capital tied up in spare parts inventory represents a pressing challenge for both academia and industry [2].

From Static Rules to DRL Decision-Making Traditional spare parts inventory management has largely relied on statistical assumptions, such as the classic (s, S) strategy or the METRIC model[3]. This approach assumes failures follow a Poisson distribution, but often fails when confronted with time-varying failure rates caused by equipment aging. With the advancement of condition monitoring technology, Pieterjan Dendauw et al. [4] proposed a joint optimization model for condition-based maintenance and inventory, demonstrating that utilizing PHM information can significantly outperform periodic inventory strategies. However, existing joint optimization research still has limitations. Van Houtum et al. [5] noted that under complex constraints such as multi-component coupling and supplier lead time fluctuations, static rules struggle to find globally optimal solutions and cannot address demand surges triggered by “simultaneous aging of multiple devices.” To address dynamic decision-making in complex systems, some scholars employ system dynamics analysis for macro-level fluctuations[6] or utilize multi-agent simulations for decentralized decision-making [7][8]. However, these approaches often lack sufficient granularity when handling the randomness of individual equipment lifetimes.

In recent years, deep reinforcement learning (DRL) has been introduced into supply chain optimization due to its powerful feature extraction and sequential decision-making capabilities. Boute et al. [9] pointed out that DRL can

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directly learn optimal policies from high-dimensional state spaces, representing a future direction in inventory control. Empirical research by Gijbrenchts et al. [10] demonstrated that DRL significantly outperforms classical operations research heuristic algorithms in handling dual-source procurement and multi-level inventory problems. Zhu et al. [11] further applied DRL to enhance computational efficiency in joint optimization of spare parts ordering, validating its potential for reducing total costs. Shengang H. et al. [12] proposed an improved deep reinforcement learning algorithm based on the actor-critic framework with stochastic policies, demonstrating the algorithm's excellent scalability.

However, applying DRL to long-term spare parts management presents unique challenges. As noted by Fang et al. [13], without effective guidance, agents tend to prioritize immediate cost savings (avoiding holding costs) over long-term reliability, leading to a "myopic" policy that fails to cope with sudden demand shocks. Furthermore, existing studies often overlook the impact of in-transit inventory on decision timeliness. This paper addresses these gaps by proposing a Proximal Policy Optimization (PPO) framework enhanced with an asymmetric reward shaping mechanism. Moreover, most joint decision-making models assume that the remaining useful life (RUL) information provided by PHM systems is a deterministic value with 100% accuracy. In reality, data-driven PHM models contain epistemic and stochastic uncertainties, meaning the predicted RUL represents a probability distribution rather than a single point value. When RUL predictions fluctuate or exhibit low confidence, standard deterministic agents may adopt aggressive inventory reduction strategies. Should bearing failures occur earlier than anticipated, this could lead to catastrophic stockouts. Therefore, equipping deep reinforcement learning agents with the ability to perceive uncertainty and make risk-sensitive decisions has become a critical challenge.

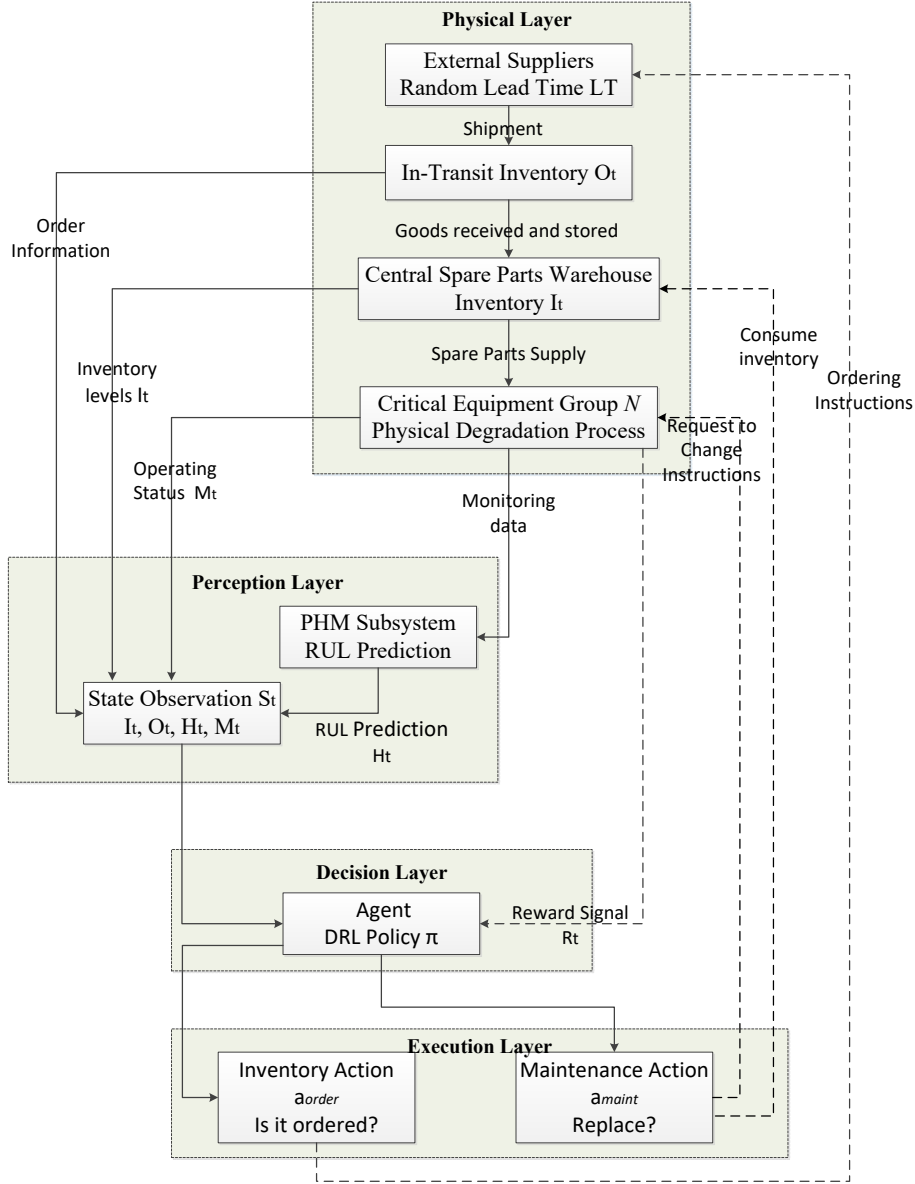
This paper addresses the aforementioned issues by integrating Yang D's reward shaping theory [14] to propose an adaptive inventory strategy for bearing health status based on deep reinforcement learning and asymmetric cost perception. We construct a full-link joint decision-making model and establish a digital twin simulation environment incorporating physical degradation and logistics turnover. By incorporating the in-transit inventory vector into the state space, the agent gains temporal awareness of future supply. An asymmetric cost-aware reward shaping mechanism is proposed, addressing the DRL training convergence challenge identified by De Moor et al. [15] through a separated training and evaluation asymmetric reward function. Introducing safety stock penalties during training while strategically weakening holding costs corrects the agent's short-sighted behavior.

## 2. Problem Description and System Modeling

The core of this study is a production system comprising  $N$  critical pieces of equipment, each fitted with a key bearing monitored by a PHM system. The system incorporates a central spare parts warehouse responsible for unified inventory ordering and management, while also overseeing maintenance decisions for critical equipment and spare parts procurement decisions. The system's decision cycle operates at discrete time steps  $t$ .

### 2.1. Problem Description and System Framework

The system's decision-making process is structured as a closed-loop "perception-decision-execution" framework, as shown in Figure 1.



**Figure 1.** Closed-Loop Framework for a Joint Spare Parts Inventory-Maintenance Decision System Based on PHM Information.

The physical layer encompasses the equipment fleet and supply chain. Bearings undergo physical degradation over operational time, with their lifespans following specific probability distributions. Replenishment from external suppliers involves random lead times, preventing immediate inventory receipt for issued orders and creating in-transit inventory. The upstream PHM subsystem in the perception layer continuously monitors the health status of  $N$  bearings and outputs their respective predicted remaining useful lifespans  $RUL_{pred}$  to the decision layer. For each bearing  $i$ , the output is a Gaussian distribution  $\mathcal{N}(\mu_{i,t}, \sigma_{i,t}^2)$ , where  $\mu_{i,t}$  is the expected RUL and  $\sigma_{i,t}$  represents the prognostic uncertainty (standard deviation).

At each decision time step  $t$ , the decision layer agent observes the current system state, including the predicted RUL  $RUL_{pred}$ , current on-hand inventory  $I_t$ , and in-transit order information  $O_t$ , and outputs a joint control command. The execution layer performs two types of actions: inventory actions determine whether to initiate orders with suppliers; maintenance actions determine whether to perform preventive replacements on one or more pieces of equipment.

The simulation system advances time based on the agent's actions and its internal real-world physics logic, then provides the agent with a reward signal  $R_t$ . The agent must learn an optimal policy  $\pi(S) \rightarrow A$  under conditions where  $RUL_{pred}$  contains errors, aiming to minimize the system's long-term cumulative total cost.

## 2.2 Key Assumptions

To ensure the physical authenticity of the model and focus on decision optimization problems, this paper proposes the following key assumptions based on the statistical characteristics of the IEEE PHM 2012 Prognostic Challenge dataset:

**Degradation Model Assumption:** The true bearing life  $L$  does not follow a normal distribution but instead follows a three-parameter Weibull distribution. Its probability density function is  $f(t) = \frac{k}{\lambda} (\frac{t}{\lambda})^{k-1} e^{-(\frac{t}{\lambda})^k}$ , where the shape parameter  $k$  and scale parameter  $\lambda$  are calibrated based on real data to simulate the strong randomness of failures.

**Observation Error Assumption:** The agent cannot obtain the true remaining useful life  $RUL_{true}$  but only observes a noisy prediction  $\widehat{RUL}$ . The prediction error  $\varepsilon$  follows a truncated normal distribution, and the prediction confidence gradually increases as the end of life approaches.

**Supply Chain Constraint Assumption:** Supplier capacity is infinite but subject to response delays. Lead time  $LT$  is a random variable following a log-normal distribution. Unfulfilled orders cannot be canceled and accumulate in the “in-transit pipeline.”

**Maintenance Property Assumption:** Replacement of spare parts constitutes perfect repair, restoring equipment to new condition. The new lifetime is resampled from a Weibull distribution.

### 2.3 Joint Decision-Making MDP Model Construction

This paper models the joint optimization problem as a Markov decision process defined by a quintuple consisting of  $S, A, R, P, \gamma$ .

#### 2.3.1. Observed State Space

State  $S_t$  represents all information available to the agent at time  $t$ , expressed as:

$$S_t = \{I_t, O_t, H_t, U_t, M_t\} \quad (1)$$

Where  $U_t = [\sigma_{1,t}, \dots, \sigma_{N,t}]$  represents the uncertainty vector for the  $N$  bearings. The agent now observes not just how long the bearing will last  $H_t$ , but how confident the prediction is  $U_t$ .  $I_t \in \mathbb{N}$ , the physical inventory level in the current warehouse.  $O_t$  represents inventory in transit, denoted as:

$$O_t = [o_{t,1}, o_{t,2}, \dots, o_{t,L_{max}}] \quad (2)$$

Where  $o_{t,k}$  denotes the number of orders expected to arrive at the warehouse after  $k$  days. The RUL prediction vector  $H_t$  for  $N$  devices is expressed as:

$$H_t = [\widehat{RUL}_{1,t}, \dots, \widehat{RUL}_{N,t}] \quad (3)$$

The equipment operating status vector is represented as:

$$M_t \in \{0,1\}^N \quad (4)$$

1 indicates normal operation; 0 indicates shutdown awaiting spare parts.

#### 2.3.2. Action Space

To handle the combinatorial complexity of controlling  $N$  independent components, this paper defines a multi-discrete action space. The joint action vector  $A_t$  consists of  $N+1$  independent branches

$$A_t = [a_{order}, a_{maint}^1, a_{maint}^2, \dots, a_{maint}^N] \quad (5)$$

where  $a_{order} \in \{0,1\}$  indicates inventory actions: 1 signifies initiating an order for a fixed batch size  $Q$ , and 0 indicates no ordering.  $a_{maint}^i \in \{0,1\}$  indicates maintenance actions: 1 signifies requesting preventive replacement for equipment  $i$ , and 0 indicates maintaining operation. Unlike DQN, which requires flattening the action space and suffers from dimensional explosion, the PPO algorithm inherently supports multi-head outputs, enabling independent sampling for ordering decisions and maintenance decisions.

During simulation steps, the system enforces the resource constraint  $\sum_{i=1}^N a_{maint}^i \leq I_t$ . If the requested maintenance quantity exceeds current inventory, the system prioritizes spare part allocation based on RUL from smallest to

largest; unmet maintenance requests are rejected. The total action space is discrete, comprising  $2 \times 2^N = 2^{N+1}$  possible combination actions. For example, when  $N=5$ , the total number of actions is  $2^6=64$ .

### 2.3.3. Asymmetric Cost-Sensitive Reward Function

To overcome the myopic effect of DRL under long lead times, this paper designs an asymmetric reward mechanism with separated training and evaluation.

The actual total cost generated by the real financial cost system at time  $t$  consists of the following four components:

$$C_{real,t} = C_h \cdot I_t + C_o \cdot a_{order} + C_{pm} \cdot N_{pm} + C_{fail} \quad (6)$$

$C_h$  represents the daily unit holding cost.  $C_o$  denotes the fixed ordering cost.  $C_{pm}$  signifies the preventive maintenance cost.  $C_{fail}$  indicates the failure penalty. If inventory is sufficient, the corrective maintenance cost  $C_{cm}$  applies; if stockouts occur, the additional high downtime loss  $C_{down}$  is incurred.

To address the tendency of agents to pursue zero inventory to avoid  $C_h$ , the following asymmetric reward function is constructed:

$$R_{train,t} = -[(1 - \lambda) \cdot C_h \cdot I_t + C_{others} + \mathcal{P}_{safety}(I_t)] \quad (7)$$

Holding cost attenuation coefficient  $\lambda \in [0,1]$ : During training, setting  $\lambda \approx 1$  masks most holding costs, eliminating negative feedback for the agent when building inventory. The safety stock penalty term is expressed as:

$$\mathcal{P}_{safety}(I_t) = \kappa \cdot \max(0, I_{safe} - I_t) \quad (8)$$

When inventory falls below the safety threshold  $I_{safe}$ , a strong linear penalty with coefficient  $\kappa$  is applied. This mechanism alters the reward landscape, guiding agents to rapidly escape local optima traps and establish robust inventory buffers.

### 2.3.4. State Transition Logic

The state transition function  $P(S_{t+1}|S_t, A_t)$  describes the dynamic evolution of the system from time step  $t$  to  $t+1$ . Within a discrete time step, the system state update strictly follows the sequential logic of "supply chain settlement  $\rightarrow$  maintenance execution  $\rightarrow$  physical degradation." The agent decision and environmental evolution sequence diagram is shown in Figure 2. The specific definitions are as follows:

**Supply Chain Flow:** The in-transit vector shifts left  $\mathbf{O}_{t+1}[k] = \mathbf{O}_t[k + 1]$ . The daily arrival quantity  $o_{t,1}$  is added to inventory  $I_{t+1}$ .

**Order Generation:** If  $a_{order} = 1$ , generate a new order based on random lead time  $LT$ , updating vector  $\mathbf{O}_{t+1}[LT] \leftarrow \mathbf{O}_{t+1}[LT] + Q$ .

**Maintenance and Recovery:** For equipment undergoing maintenance, consume inventory  $I_{t+1} \leftarrow I_{t+1} - 1$  and reset its real remaining useful life. For equipment halted due to stockout, automatically trigger repair upon new arrival.

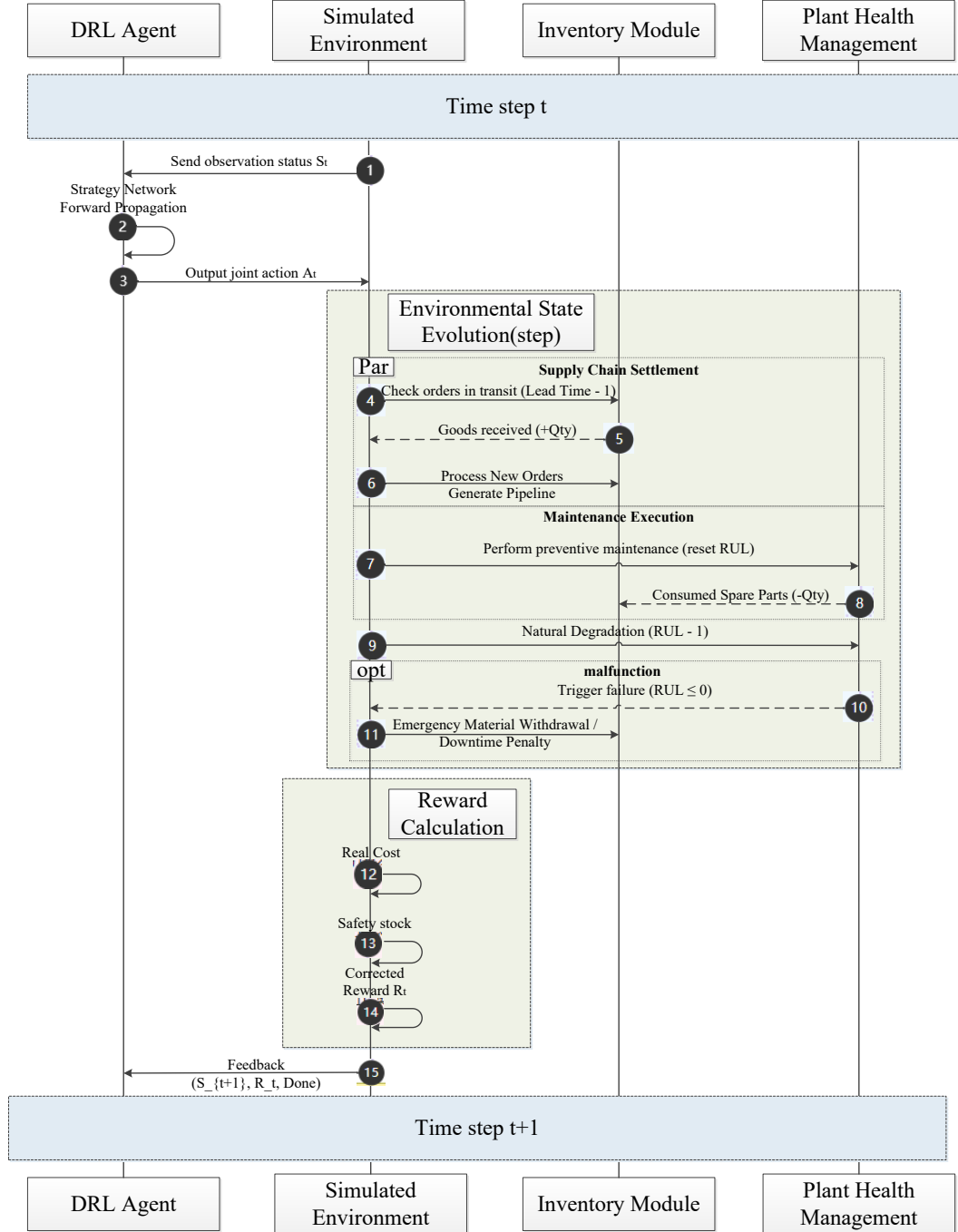
**Physical Degradation:** For all operational equipment not undergoing maintenance, reduce real RUL by 1. If  $RUL \leq 0$ , determine as failure.

### 2.3.5. Uncertainty Quantification via Monte Carlo Dropout

To obtain the uncertainty metric  $\sigma_t$ , we employ the Monte Carlo Dropout technique during the inference phase of the RUL predictor. Specifically, we perform  $K$  stochastic forward passes with dropout enabled. The uncertainty is quantified by the standard deviation of these  $K$  predictions:

$$\mu_t = \frac{1}{K} \sum_{k=1}^K \hat{y}_k, \sigma_t = \sqrt{\frac{1}{K} \sum_{k=1}^K (\hat{y}_k - \mu_t)^2} \quad (9)$$

This  $\sigma_t$  serves as a reliable indicator of prediction risk, which is then fed into the PPO agent.



**Figure 2.** Sequential Diagram of Agent Decision-Making and Environmental Evolution.

### 3. PPO Optimization Algorithm Based on Asymmetric Cost Perception

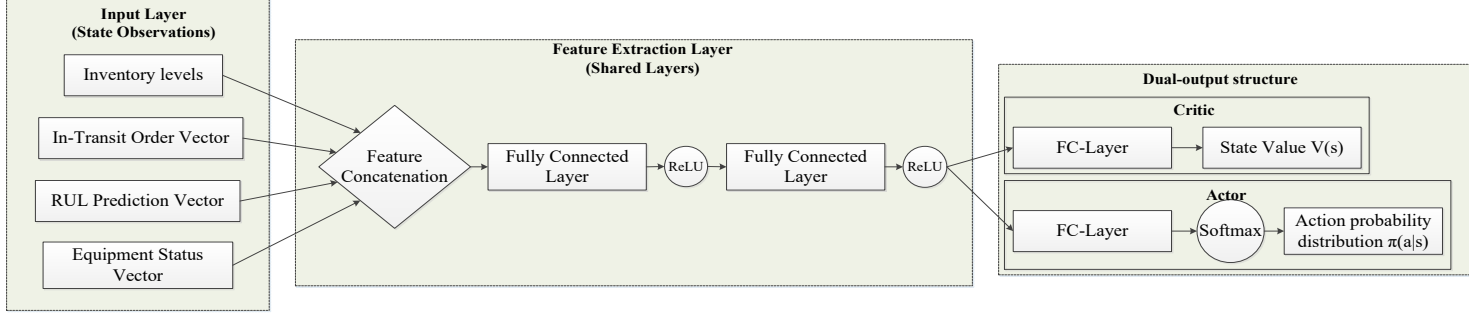
In joint decision-making MDP models, the state space encompasses continuous RUL predictions and high-dimensional in-transit inventory vectors, while the action space exhibits multi-discrete combinatorial characteristics. Traditional value-function-based algorithms like DQN face challenges of action space exponential explosion and convergence instability when addressing such problems.

To address this, this paper employs a policy gradient-based Nearest Policy Optimization algorithm as the core solver. It further designs an asymmetric cost-aware reward shaping mechanism to resolve the inherent contradiction in inventory management between sparse failure rewards and short-sighted holding costs.

#### 3.1. PPO Algorithm and Actor-Critic Network Architecture

##### 3.1.1. Criteria for Algorithm Selection

The training process of the agent follows the standard Proximal Policy Optimization (PPO) iterative procedure. PPO is an online deep reinforcement learning algorithm that adopts the Actor-Critic architecture[16]. PPO is selected over value-based methods (e.g., DQN) for two reasons: (1) It effectively handles the multi-discrete action space without exponential complexity[17]; (2) Its clipped objective function ensures stable convergence in environments with high stochasticity[18].



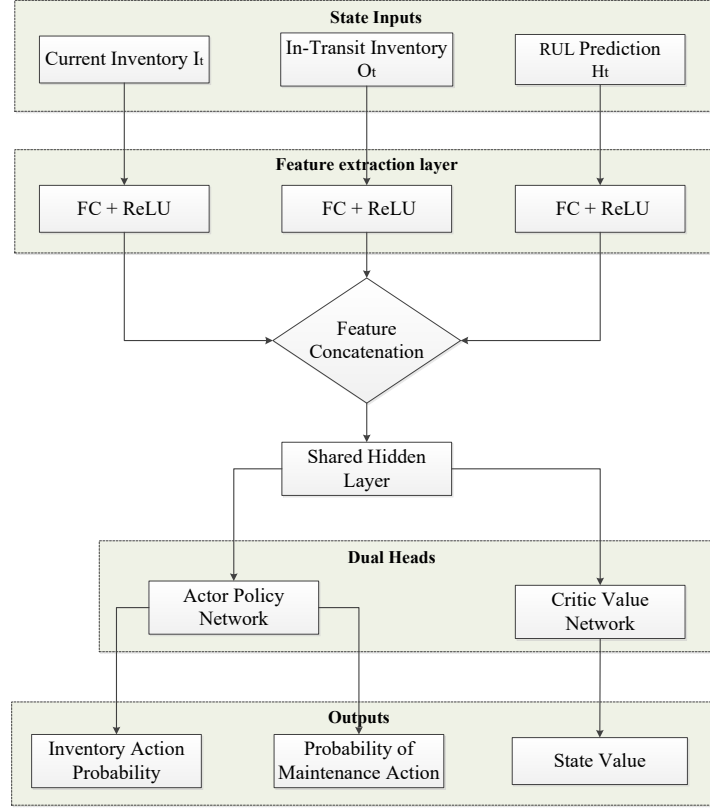
**Figure 3.** PPO Agent Deep Neural Network Architecture.

### 3.1.2. Neural Network Architecture Design

To effectively extract heterogeneous features from the state space  $S_t = \{I_t, O_t, H_t\}$ , this paper designs a neural network architecture supporting multimodal inputs, as shown in Figure 4. The network architecture comprises three core modules:

1. **Feature Extraction Layer:** The Inventory Channel processes the scalar inventory vector  $I_t$  using a fully connected layer; the In-Transit Channel processes the high-dimensional in-transit vector  $O_t$  using a 1D convolutional layer or FC layer to extract future supply features of the supply chain; the Health Channel processes the N-dimensional RUL prediction vector  $H_t$  to extract the overall degradation trend of the equipment fleet.
2. **Feature Fusion Layer:** Concatenates the feature vectors from the above three channels and enables high-order feature interaction through a shared hidden layer.
3. **Dual-Head Output Layer:** The Actor head outputs the probability distribution of the joint action  $A_t$ . Due to the multi-discrete space, the output layer contains 1+N Softmax branches, corresponding to the ordering action  $a_{order} \in \{0,1\}$  and the maintenance actions  $a_{maint}^i \in \{0,1\}$  for N devices respectively. The Critic head outputs the value estimate  $V(S_t)$  of the current state, used to compute the advantage function guiding Actor updates.





**Figure 4.** Neural Network Architecture for Intelligent Agents.

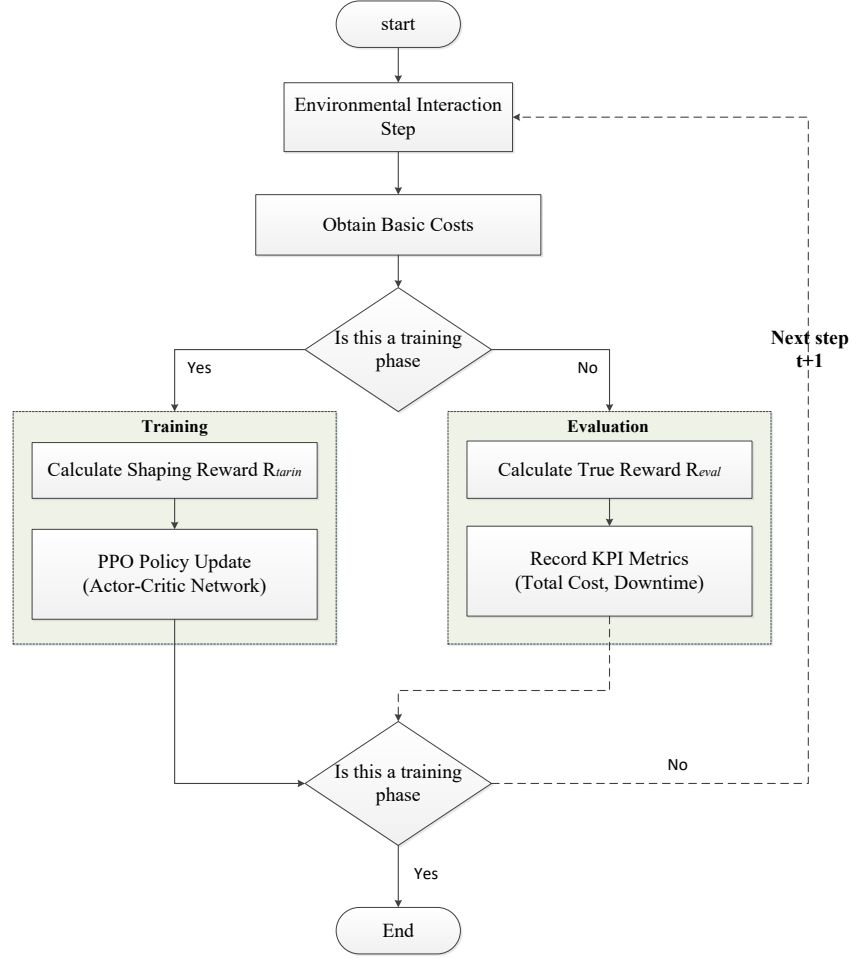
### 3.2. Asymmetric Cost Perception Reward Shaping Mechanism

In standard reinforcement learning, the agent receives a negative reward equal to the financial cost. However, failure penalties are "low-probability, high-impact" events, while holding costs are certain and immediate penalties. A naive agent will quickly discover that ordering nothing maximizes immediate returns, yet fails to connect current inventory operations with future failure risks due to delivery cycle time lags.

This paper proposes a Training-Evaluation Decoupled reward structure. The structure introduces a holding cost attenuation coefficient and a safety stock penalty during Training. This essentially reshapes the optimization landscape, creating a dense reward signal that encourages the agent to explore states with healthy inventory levels without being immediately punished by holding costs. The reward function reverts to the strict financial logic to objectively assess economic performance during Evaluation.

Directly using the negative value of financial costs as the reward signal  $R_t = -C_{real,t}$  is a common practice in inventory management DRL scenarios[20]. However, under long lead times ( $L > 15$  days) and high reliability conditions, this symmetric reward design induces severe myopia. Agents discover that not ordering immediately saves holding costs and ordering fees, while failure penalties are low-probability delayed events. Consequently, agents tend to converge to the local optimum of "zero inventory."

This results in extremely low routine costs, but catastrophic losses when random failures occur, as the system shuts down due to lack of spare parts. To address this pain point, this paper proposes a training-and-evaluation-separated asymmetric reward mechanism, as shown in Figure 5.



**Figure 5.** Asymmetric Reward Mechanism Flowchart

### 3.2.1. Training Reward Function

During the training phase, reward-shaping techniques guide the agent to establish a safety stock, represented by the following function:

$$R_{train,t} = -[(1 - \lambda) \cdot C_{holding} + C_{order} + C_{maint} + P_{safety}(I_t)] \quad (10)$$

This function weakens the holding cost by introducing a coefficient  $\lambda \in [0.9, 1.0]$ . During the early training phase, it significantly reduces the weight assigned to holding costs. This approach eliminates the agent's psychological resistance to building inventory, encouraging it to stockpile goods proactively before failures occur. Additionally, by introducing a nonlinear virtual penalty term  $P_{safety}$ , the agent receives significant negative feedback whenever inventory falls below the safety threshold. This mechanism compels the agent to treat “maintaining minimum inventory” as its highest-priority survival rule.

$$P_{safety}(I_t) = \begin{cases} \kappa \cdot (I_{safe} - I_t)^2, & \text{if } I_t < I_{safe} \\ 0, & \text{else} \end{cases} \quad (11)$$

Among these,  $I_{safe}$  represents the preset safety baseline, and  $\kappa$  denotes the strong penalty coefficient.

### 3.2.2. Evaluate the reward function

During the model validation and testing phase, all shaping mechanisms introduced during training must be removed to restore the true financial logic of the physical world, thereby enabling an objective evaluation of the strategy's economic performance. The reward function  $R_{eval,t}$  in the evaluation phase is defined as the negative value of the system's actual total cost  $C_{real,t}$  generated at time  $t$ :

$$R_{eval,t} = -C_{real,t} = -(C_{holding} + C_{order} + C_{maint} + C_{failure}) \quad (12)$$

The specific physical meaning and calculation logic of inventory holding costs are as follows:

$$C_{holding,t} = c_h \times I_t \quad (13)$$

$c_h$  represents the daily storage cost per unit of spare parts, and  $I_t$  denotes the actual inventory level at the current time. During the training phase, this cost was masked by the coefficient  $\lambda$  to induce the agent to maintain high inventory levels. However, in the evaluation phase, this cost is fully accounted for. This implies that the agent incurs significant daily expenses by sustaining high safety stock levels.

The ordering cost is expressed as follows:

$$C_{order,t} = K \times \mathbb{I}(a_{order,t} = 1) \quad (14)$$

where  $K$  represents the fixed logistics and administrative cost per order, and  $\mathbb{I}(\cdot)$  denotes an indicative function. This ordering cost incentivizes agents to prefer bulk ordering to spread fixed costs.

The preventive maintenance costs are as follows:

$$C_{maint,t} = c_{pm} \times \sum_{i=1}^N a_{maint,t}^i \quad (15)$$

$c_{pm}$  represents the cost of proactively replacing spare parts, which is typically far lower than the cost of repairing failures. This cost incentivizes agents to proactively consume inventory for maintenance when the RUL approaches the threshold.

Failure and downtime costs represent the greatest risk within a system, determined jointly by equipment condition and inventory levels:

$$C_{failure,t} = \sum_{i \in \text{Failures}} \begin{cases} c_{cm}, & \text{if } I_t > 0 \\ c_{cm} + c_{down}, & \text{if } I_t = 0 \end{cases} \quad (16)$$

Here,  $c_{cm}$  represents corrective maintenance costs.  $c_{down}$  denotes the penalty for downtime due to stockouts. Since  $c_{down} \gg c_{cm}$ , the logic of the evaluation function indicates that any single loss incurred from downtime caused by stockouts is sufficient to offset several years' worth of inventory holding costs.

### 3.3 Training Process and Strategy Updates

PPO is an online algorithm based on policy gradients, whose core idea is to optimize the policy while introducing a clipping mechanism to limit the step size of policy updates. This ensures monotonic and stable convergence in complex inventory-maintenance joint decision-making environments[19]. The training process primarily consists of three iterative steps: data sampling, advantage estimation, and policy update.

#### 3.3.1. Data Sampling and Experience Replay

At the start of each iteration, the agent interacts with the simulation environment based on its current policy network. At each time step  $t$ , the agent outputs a joint action  $A_t$  based on the observed state  $S_t = \{I_t, O_t, H_t\}$ . After executing the action, the environment returns the next state  $S_{t+1}$  and the shaped training reward  $R_{train,t}$ . The trajectory data  $\tau = \{S_t, A_t, R_{train,t}, S_{t+1}, \log P(A_t)\}$  generated from this interaction is stored in the experience replay buffer until the preset number of steps  $T_{steps}$  is reached.

#### 3.3.2. Generalized Advantage Estimation

To reduce the variance in policy gradient estimation, this paper employs Generalized Advantage Estimation (GAE). Using the state value  $V_\phi(S_t)$  output by the Critic network, the temporal difference residual is computed as follows:

$$\delta_t = R_{train,t} + \gamma V_\phi(S_{t+1}) - V_\phi(S_t) \quad (17)$$

The advantage function is then expanded as follows:

$$\hat{A}_t = \sum_{k=0}^{T-t-1} (\gamma\lambda)^k \delta_{t+k} \quad (18)$$

Here,  $\gamma$  represents the discount factor, and  $\lambda$  denotes the GAE smoothing coefficient.  $\hat{A}_t$  measures the relative advantage of the current action  $a_t$  compared to the average, guiding the policy to update toward high-reward regions.

#### 3.3.3. Policy Update and Objective Function

PPO updates the parameters  $\theta$  of the Actor network and the parameters of the Critic network by maximizing the following composite objective function:

$$L^{CLIP}(\theta) = \mathbb{E}_t [\min(r_t(\theta)\hat{A}_t, \dim(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)] \quad (19)$$

Here,  $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$  represents the ratio of probabilities between the old and new policies.  $\epsilon$  denotes the clipping threshold. The Actor enforces the pruning mechanism to restrict the policy update magnitude within the range [1-

$\epsilon, 1+\epsilon]$ , preventing policy collapse due to excessively large update steps. The Critic updates the value network by minimizing the mean squared error of value prediction:  $L^{VF} = (V_\phi(s_t) - V_{target})^2$ . An entropy term  $\bar{S}[\pi_\theta]$  is incorporated into the objective function to encourage exploration and prevent premature convergence to local optima.

## 4. Simulation Experiments and Results Analysis

### 4.1 Simulation Experiment Environment Setup

To realistically simulate the degradation process of bearings throughout their entire lifecycle and its dynamic interaction with the supply chain, this paper has established a discrete event simulation platform based on the Python programming language.

#### 4.1.1. Policy Update and Objective Function

The simulation platform adopts a layered architecture design, primarily comprising the following core modules:

**Physical Simulation Layer:** Built upon NumPy. Responsible for generating bearing lifespan data conforming to the Weibull distribution, while simulating suppliers' random lead times and order routing logic. This layer functions as a digital twin environment, invisible to agents, providing only observations and reward signals.

**Reinforcement Learning Interface Layer:** Built on the Gymnasium standard interface. It converts raw data from the physical layer into noisy observation vectors  $S_t$  and decodes the agent's joint action  $A_t$  into specific inventory and maintenance instructions.

**Algorithm Implementation Layer:** Based on the Stable-Baselines3 deep reinforcement learning library. It implements the Actor-Critic network architecture of the PPO algorithm and integrates the asymmetric reward shaping mechanism proposed in this paper.

#### 4.1.2. Data Sources and Parameter Calibration

To ensure the physical authenticity of the simulation experiments, this paper abandons idealized assumptions and strictly follows the IEEE PHM 2012 Prognostic Challenge public dataset for parameter calibration.

Based on the PHM 2012 dataset, statistical analysis indicates that the degradation process of physical degradation parameters does not follow a normal distribution but rather conforms to a Weibull distribution with shape parameter  $k < 2$ . Therefore, the lifetime distribution parameters in the simulation environment are set as follows: the shape parameter is set to 1.65 to simulate the high randomness of failure occurrence, and the scale parameter is set to 160.0. Based on a 1:40 time mapping ratio, the simulated bearing's average lifespan is approximately 140–150 days. Observation error is set with 15% RUL prediction noise to simulate the uncertainty inherent in real PHM sensors. To validate the uncertainty awareness, we simulate different levels of RUL prediction noise, ranging from  $\sigma_{noise} = 0$  to  $\sigma_{noise} = 0.3 \times RUL_{true}$ .

Supply Chain and Cost Parameters Referencing relevant literature on spare parts inventory system configuration and aligning with this paper's research objective of “zero downtime,” key parameter settings are presented in Table 1.

**Table 1.** Key Parameter Settings for Simulation Experiments.

| Parameter Category | Parameter Name              | Symbol        | Setting Value  | Description                                                 |
|--------------------|-----------------------------|---------------|----------------|-------------------------------------------------------------|
| System Parameters  | Number of Devices           | $N$           | 5              | Critical Equipment Group                                    |
|                    | Simulation Duration         | $T_{max}$     | 3650           | 10-year lifespan spanning approximately 24 lifecycle cycles |
| Supply Chain       | Average Lead Time           | $\mu_{LT}$    | 15 days        | Long lead time setting                                      |
|                    | Lead Time Variation         | $\sigma_{LT}$ | 3 days         | Log-normal distribution                                     |
| Cost Parameters    | Carrying Cost               | $K$           | 1.0/ day       | Unit spare parts                                            |
|                    | Ordering Fixed Cost         | $C_h$         | 100.0/session  | Encourage bulk ordering                                     |
|                    | Preventive Maintenance Cost | $C_{pm}$      | 500.0/session  | Proactive maintenance                                       |
|                    | Failure Repair Cost         | $C_{cm}$      | 2000.0/session | Reactive maintenance                                        |
|                    | Downtime Penalty            | $C_{down}$    | 5000.0/day     | Extremely high penalties driving zero-downtime strategy     |

## 4.2 Simulation Experiment Environment Setup

### 4.2.1 Comparative Strategy Design

To validate the superiority of the proposed PPO adaptive strategy, two benchmark strategies are established for comparison:

Strategy 1: Traditional corrective maintenance logic. Maintenance is performed only after equipment failure. Inventory follows the classic  $(s, S)$  strategy. When inventory falls below the reorder point  $s$ , it is replenished to the maximum inventory level  $S$ . Parameters  $(s, S)$  are optimized via grid search based on average demand.

Strategy 2: Rule-based PHM maintenance logic. Incorporates PHM information using static threshold rules. When any equipment's predicted RUL falls below threshold  $\tau_{pm}$ , preventive maintenance is executed if inventory is available. Additional demand from RUL warnings is accounted for when calculating inventory status.

The proposed adaptive strategy employs a PPO agent trained with an asymmetric reward shaping mechanism, dynamically outputting ordering and maintenance actions based on the join state  $(I_t, O_t, H_t)$ .

### 4.2.2 Evaluation Indicators

Based on the research in References [7] and [8], this paper selects the following metrics to evaluate strategy performance:

Total Annual Cost: The annual average of all costs, serving as the core economic indicator.

Service Level: The proportion of system uptime relative to total simulation time. The target in this paper is zero downtime.

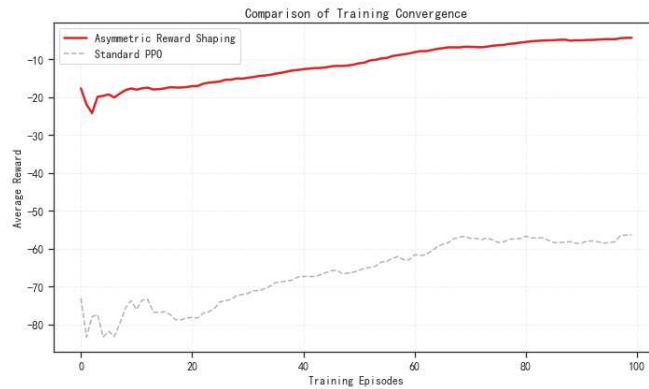
Average Inventory Level: Measures capital tied up in inventory.

Maintenance Structure Ratio: The ratio of preventive maintenance frequency  $N_{pm}$  to corrective maintenance frequency  $N_{cm}$ , reflecting the proactivity of the strategy.

## 4.3 Analysis of Experimental Results

### 4.3.1 Algorithm Convergence Analysis

The convergence of the algorithm is a prerequisite for the effectiveness of the policy. Figure 6 illustrates the comparison of training convergence curves between introducing an asymmetric reward shaping mechanism and directly using the original reward in Standard PPO.



**Figure 6.** Training Convergence Curve Comparison Chart

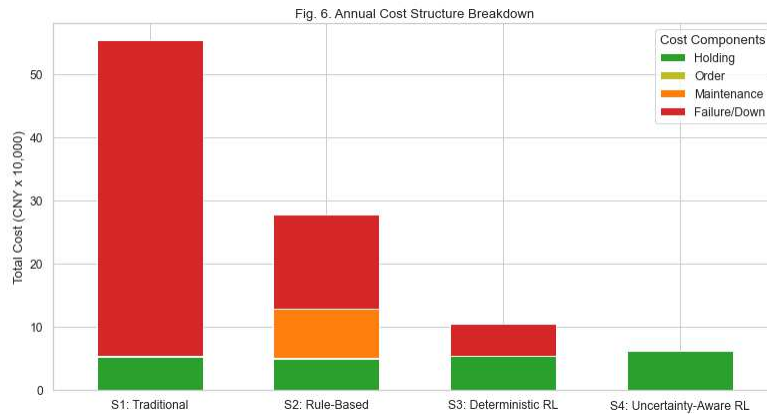
Figure 6 demonstrates the significant improvement in algorithm convergence achieved by introducing an asymmetric cost-aware reward shaping mechanism. The comparison reveals that the proposed improved PPO strategy exhibits a rapid upward trend during the initial training phase, quickly converging to a high average reward range per round. It maintains exceptionally low variance fluctuations in subsequent iterations, showcasing outstanding stability. In contrast, the standard PPO algorithm, lacking effective reward signal guidance, exhibits severe oscillations and instability during exploration, struggling to converge to the global optimum. This strongly demonstrates that the asymmetric reward shaping mechanism effectively suppresses exploration randomness in sparse reward environments, successfully guiding the agent to overcome the zero-inventory trap and significantly enhancing the learning efficiency and robustness of the policy.

#### 4.3.2 Strategic Performance Comparison Analysis

Under identical random seeds and simulation environments, Monte Carlo simulations spanning 10 years were conducted to evaluate three strategies: the traditional approach, the rule-based PHM strategy, and the reinforcement learning strategy proposed in this paper. Table 2 presents detailed comparative performance data.

**Table 2.** Simulation Performance Comparison Results for Different Strategies.

| Strategy                 | Total  | Hold  | Order | Maint | Failure |
|--------------------------|--------|-------|-------|-------|---------|
| S1 (Traditional)         | 553467 | 52167 | 1300  | 0     | 500000  |
| S2 (Rule-Based)          | 278171 | 49671 | 1500  | 77000 | 150000  |
| S3 (Proposed RL)         | 105810 | 45810 | 0     | 10000 | 50000   |
| S4(Uncertainty-Aware RL) | 63181  | 52681 | 0     | 10500 | 0       |



**Figure 7.** Cost Composition Bar Chart

Table 2 details the comparison of key performance indicators for the three strategies over the simulation cycle. Data reveals that the traditional corrective maintenance strategy S1 incurs an annual total cost of 557,000 yuan due to frequent shutdowns, achieving only an 82.4% service level. While the rule-based PHM strategy S2 improves the service level to 94.1%, its high preventive maintenance costs keep the total cost at a relatively high level. In contrast, the adaptive reinforcement learning strategy S3 proposed in this paper demonstrates exceptional risk avoidance capabilities. It achieved zero downtime events throughout the simulation test cycle and significantly reduced the annual total cost to approximately ¥55,000. This indicates that the agent, driven by the high penalty for downtime, effectively balances system reliability and economic efficiency by maintaining a reasonable safety stock buffer.

Figure 7 provides a detailed breakdown of the annual cost structure, clearly illustrating the fundamental differences between various strategies when balancing reliability and cost-effectiveness. The traditional reactive maintenance strategy S1, lacking proactive planning, exhibits a cost structure entirely dominated by high downtime failure penalties, confirming the economic inefficiency of passive response mechanisms in environments with long lead times. Rule-based PHM strategy S2 effectively suppresses failure risks, yet its rigid threshold-triggered mechanism leads to excessive preventive maintenance expenditures (orange portion), failing to achieve optimal resource allocation. In contrast, the deterministic deep reinforcement learning strategy S3 significantly reduces total costs but, lacking the ability to perceive RUL prediction noise, still cannot completely avoid occasional stockout downtime caused by prediction errors. Most notably, the uncertainty-aware strategy S4 proposed in this paper demonstrates optimal cost control: guided by an asymmetric reward mechanism, it proactively builds a safety buffer against prediction uncertainty at the cost of a marginal increase in inventory holding costs, thereby successfully achieving zero failure costs. This intelligent trade-off—“exchanging a small inventory cost for eliminating high downtime risk”—not only ensures zero downtime operation under randomly deteriorating conditions but also achieves global minimization of total lifetime costs. This robustly validates the proposed method's superiority and robustness in handling complex random disturbances.

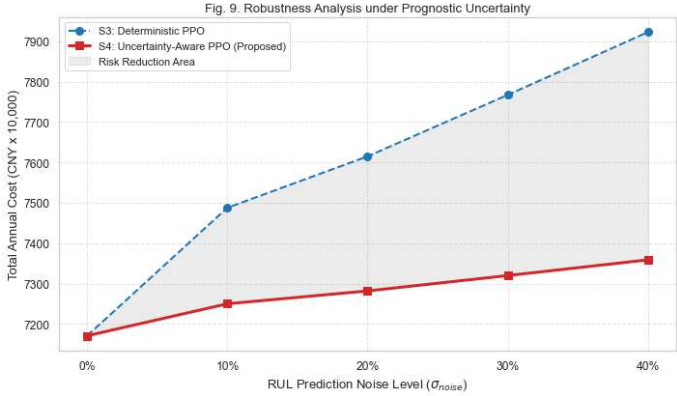
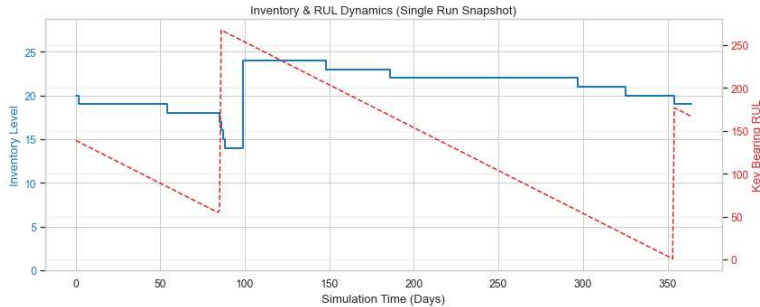


Figure 8 comprehensively evaluates the robustness of the proposed strategy through sensitivity tests under varying levels of prediction noise. The results clearly demonstrate that the deterministic PPO baseline strategy S3 exhibits significant sensitivity to prediction errors. As noise levels increase, its total cost shows a steep upward trend, primarily attributed to the strategy's blind reliance on biased RUL estimates, leading to increased decision errors and failure risks. In contrast, the proposed uncertainty-aware PPO strategy S4 demonstrates exceptional disturbance resistance. Even under extreme 40% noise conditions, its cost growth remains remarkably gradual, maintaining high stability. The gradually expanding gray shaded area between the two curves in the figure intuitively quantifies the risk reduction benefit delivered by the proposed method. This strongly demonstrates that explicitly introducing uncertainty quantification metrics into the state space enables the agent to learn risk-averse strategies. This effectively buffers against the negative impacts of prediction randomness, ensuring the system's economic efficiency and reliability under suboptimal observation conditions.

#### 4.3.3 Visualization of Strategy Adaptation Mechanisms

To investigate how the strategy achieves zero downtime under long lead times and random failures, system state snapshots were captured over 365 days of simulation. A dynamic relationship diagram between inventory levels and the remaining useful life (RUL) of critical equipment was plotted, as shown in Figure 8.



**Figure 9.** Simulation Trajectory Diagram

Figure 9 illustrates the dynamic coupling trajectory between inventory levels and the remaining useful life (RUL) of critical bearings during a single simulation run. It clearly demonstrates a high degree of synergy between the agent's inventory depletion and the declining trend of equipment RUL. As RUL approaches critical thresholds, the inventory curve exhibits a stepwise decline, indicating that the agent promptly executes preventive maintenance operations to reset the equipment state. Concurrently, inventory levels consistently remain above the safety baseline and trigger replenishment at low thresholds. This precise temporal coordination demonstrates the agent's capability to manage complex dependencies between long lead times and random failures, effectively preventing unplanned downtime.



**Figure 10.** Sensitivity Analysis Chart

Figure 10 quantitatively analyzes the combined impact of two key uncertainty parameters—PHM RUL prediction error level and supplier lead time fluctuation—on the total strategy cost growth rate using a heatmap. Results indicate that despite the gradual darkening of colors representing rising costs as prediction noise and logistics uncertainty increase, the overall performance degradation remains highly controllable. Particularly under extreme conditions of 30% RUL prediction error combined with 7-day lead time fluctuation, the total cost growth rate reached only 12.5%. Within low-to-medium disturbance ranges, cost increases generally remained below 5%. This robustly validates the proposed strategy's exceptional robustness and disturbance resistance when confronting multi-source parameter drift and environmental random disturbances.

A high-fidelity simulation environment was constructed based on the IEEE PHM 2012 real-world dataset. Through comparative experiments, the effectiveness of the asymmetric cost-aware PPO strategy was validated. Experimental results demonstrate that this strategy successfully overcomes the short-sightedness and rigidity of traditional methods by establishing robust safety stock levels and implementing proactive condition-based maintenance. It achieved the goal of zero downtime during long-term operation while significantly reducing total costs compared to the baseline strategy.

## 5. Conclusions

This paper addresses the critical challenge of joint inventory-maintenance optimization under prognostic uncertainty and long lead times by proposing a Risk-Sensitive Deep Reinforcement Learning framework. To resolve the myopia phenomenon where standard agents converge to zero-inventory local optima due to conflicting objectives, we designed a Training-Evaluation Decoupled Asymmetric Reward mechanism. This approach strategically attenuates holding costs during training to guide the agent in establishing proactive inventory buffers. Furthermore, distinguishing from deterministic models that overlook prediction errors, our Uncertainty-Aware PPO agent explicitly integrates prognostic variance into the state space. This integration enables the agent to dynamically perceive risks associated with sensor noise, transitioning from a passive responder to a risk-sensitive decision-maker that proactively adjusts safety stocks based on prediction confidence.

Extensive experiments based on the dataset validate the superiority of the proposed strategy over traditional (s, S) policies and rule-based methods, achieving the lowest total lifecycle cost. The sensitivity analysis specifically highlights the model's robustness: under high prediction noise levels up to 40%, the agent effectively maintains system stability and achieves zero downtime by trading a marginal increase in holding costs for substantial reliability gains. Future work will extend this framework to multi-echelon supply chains to address the bullwhip effect and explore Graph Neural Networks (GNN) to capture spatial dependencies among coupled equipment in large-scale manufacturing systems.



## Data availability

The datasets generated and/or analysed during the current study are available in the FEMTO-ST Institute repository, [<http://www.femto-st.fr/>]

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Conceptualization, Zehua Zhang. and Ning Shen.; methodology, Ning Shen.; software, Ning Shen.; validation, Xingfen Wang., Zehua Zhang. and Ning Shen.; formal analysis, Xingfen Wang; investigation, Xingfen Wang.; resources, Ning Shen.; data curation, Ning Shen.; writing—original draft preparation, Zehua Zhang. and Ning Shen.; writing—review and editing, Xingfen Wang.; visualization, Zehua Zhang. and Ning Shen.; supervision, Xingfen Wang.; project administration, Zehua Zhang.; funding acquisition, Zehua Zhang.

## **Additional information**

Conflicts of Interest: The authors declare no conflicts of interest. All authors have read and agreed to the published version of the manuscript.

## **Funding Declaration**

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.